Pie Tactic Predictor

Fine-tuning Experiments Report

Auto-generated by `generate_report.py`

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Contents

1 Dataset

The training corpus consists of proof tactic prediction examples extracted from the Pie language. Each example provides the current proof goal together with local and global contexts, and the model must predict the correct tactic to apply.

Table 1: Dataset statistics

Statistic	Value
Total examples	7,840
Unique theorems	1,479
Unique tactic types	11
Training split	6535
Validation split	727
Test split	578

2 Models and Hyperparameters

4 candidate models spanning 1.3B to 34B parameters were evaluated. Table 2 lists each model alongside its LoRA and training hyperparameters. 4 of 4 models have completed training with valid results.

Table 2: Model catalogue and hyperparameter configuration

Model ID	HuggingFace ID	LoRA r	α	Batch	Grad. acc.	Eff. batch	LR	Epochs	Trained
`deepseek-coder-1.3b`	`deepseek-ai/deepseek-coder-1.3b-base`	64	128	16	2	32	0.0002	5	✓
`deepseek-coder-6.7b`	`deepseek-ai/deepseek-coder-6.7b-base`	32	64	8	4	32	0.0001	5	✓
`qwen2.5-coder-1.5b`	`Qwen/Qwen2.5-Coder-1.5B`	64	128	16	2	32	0.0002	5	✓
`qwen2.5-coder-7b`	`Qwen/Qwen2.5-Coder-7B`	32	64	8	4	32	0.0001	5	✓

3 Training Results

Table 3 summarises the final training and best validation metrics for all successfully completed runs.

Table 3: Training run summary (best eval metrics across all checkpoints)

Model	Time (min)	Steps	Train loss	Train acc.	Best val. loss	Best val. acc.
`deepseek-coder-1.3b`	71.7	1025	0.0112	99.38%	0.1970	95.90%
`deepseek-coder-6.7b`	174.1	1025	0.0081	99.54%	0.1456	96.89%
`qwen2.5-coder-1.5b`	66.7	1025	0.0106	99.42%	0.2559	94.89%
`qwen2.5-coder-7b`	136.8	1025	0.0061	---	0.1490	96.48%

4 Learning Curves

Figures 1 and 2 show the per-step training loss / token accuracy and the per-checkpoint validation metrics for all models that completed training. Models with degenerate zero-accuracy behaviour are excluded from the accuracy subplots for clarity.

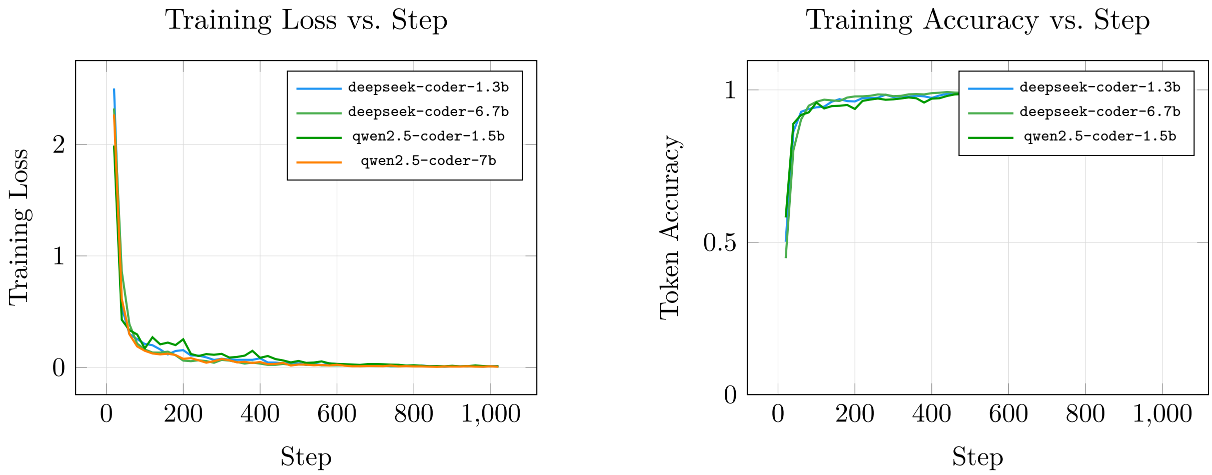

Figure 1: Training loss (left) and per-step token accuracy (right) across training steps.

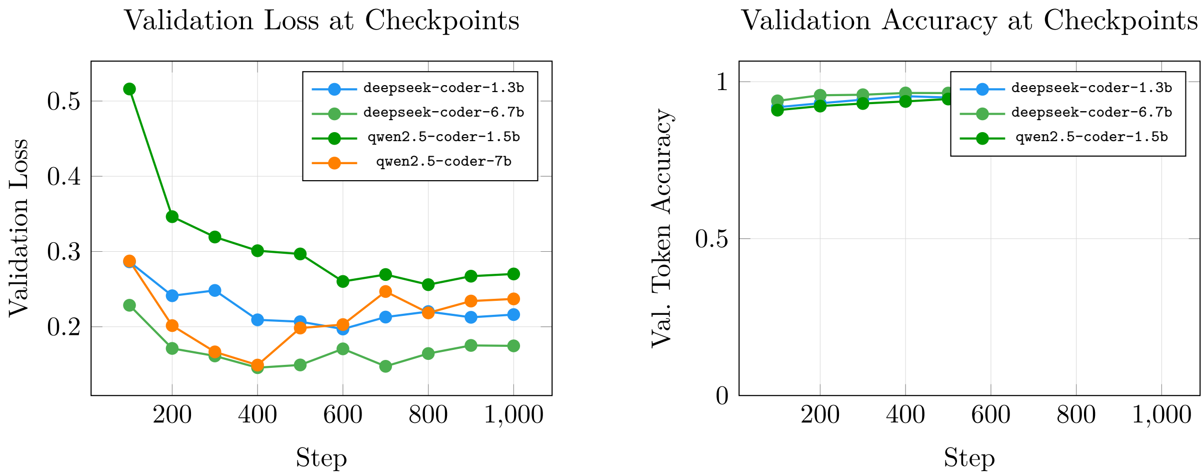

Figure 2: Validation loss (left) and validation token accuracy (right) at each evaluation checkpoint.

5 Analysis and Discussion

5.1 Completed Runs

The following models trained successfully and produced meaningful metrics: `deepseek-coder-1.3b`, `deepseek-coder-6.7b`, `qwen2.5-coder-1.5b`, `qwen2.5-coder-7b`. The best-performing model overall was `deepseek-coder-6.7b` (96.89% val. accuracy).

5.2 Compute Budget

Total GPU time across all completed runs was approximately **7.5 GPU-hours**.

A Per-Model Hyperparameter Details

A.1 `deepseek-coder-1.3b`

Parameter	Value
HuggingFace ID	`deepseek-ai/deepseek-coder-1.3b-base`
LoRA rank (r)	64
LoRA α	128
LoRA dropout	0.05
Epochs	5
Batch size (per device)	16
Gradient accumulation	2
Effective batch size	32
Learning rate	0.0002
Warmup ratio	0.05
Max sequence length	1024
Load in 4-bit	No
Load in 8-bit	No
Total training steps	1025
Total training time (min)	71.7
Final training loss	0.0112
Final training accuracy	99.38%
Best validation loss	0.1970
Best validation accuracy	95.90%

A.2 `deepseek-coder-6.7b`

Parameter	Value
HuggingFace ID	`deepseek-ai/deepseek-coder-6.7b-base`
LoRA rank (r)	32
LoRA α	64
LoRA dropout	0.05
Epochs	5
Batch size (per device)	8
Gradient accumulation	4
Effective batch size	32
Learning rate	0.0001
Warmup ratio	0.05
Max sequence length	1024
Load in 4-bit	No
Load in 8-bit	No
Total training steps	1025
Total training time (min)	174.1
Final training loss	0.0081
Final training accuracy	99.54%
Best validation loss	0.1456
Best validation accuracy	96.89%

A.3 `qwen2.5-coder-1.5b`

Parameter	Value
HuggingFace ID	`Qwen/Qwen2.5-Coder-1.5B`
LoRA rank (r)	64
LoRA α	128
LoRA dropout	0.05
Epochs	5
Batch size (per device)	16
Gradient accumulation	2
Effective batch size	32
Learning rate	0.0002
Warmup ratio	0.05
Max sequence length	1024
Load in 4-bit	No
Load in 8-bit	No
Total training steps	1025
Total training time (min)	66.7
Final training loss	0.0106
Final training accuracy	99.42%
Best validation loss	0.2559
Best validation accuracy	94.89%

A.4 `qwen2.5-coder-7b`

Parameter	Value
HuggingFace ID	`Qwen/Qwen2.5-Coder-7B`
LoRA rank (r)	32
LoRA α	64
LoRA dropout	0.05
Epochs	5
Batch size (per device)	8
Gradient accumulation	4
Effective batch size	32
Learning rate	0.0001
Warmup ratio	0.05
Max sequence length	1024
Load in 4-bit	No
Load in 8-bit	No
Total training steps	1025
Total training time (min)	136.8
Final training loss	0.0061
Final training accuracy	---
Best validation loss	0.1490
Best validation accuracy	96.48%